Advanced Macroeconomics

Problem Set 1

Business Cycles & Local Projections

Francesco Viglietta

April 3, 2026

1 Business Cycle Data

1.1

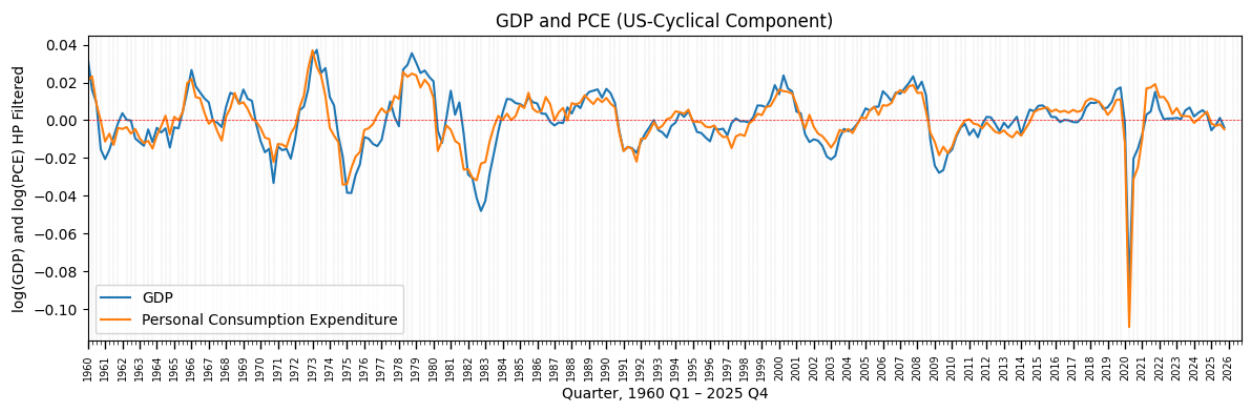

Figure 1. Cyclical Components of GDP and PCE – United States

Note: Both series are expressed in log terms and HP-filtered, $\lambda = 1600$.

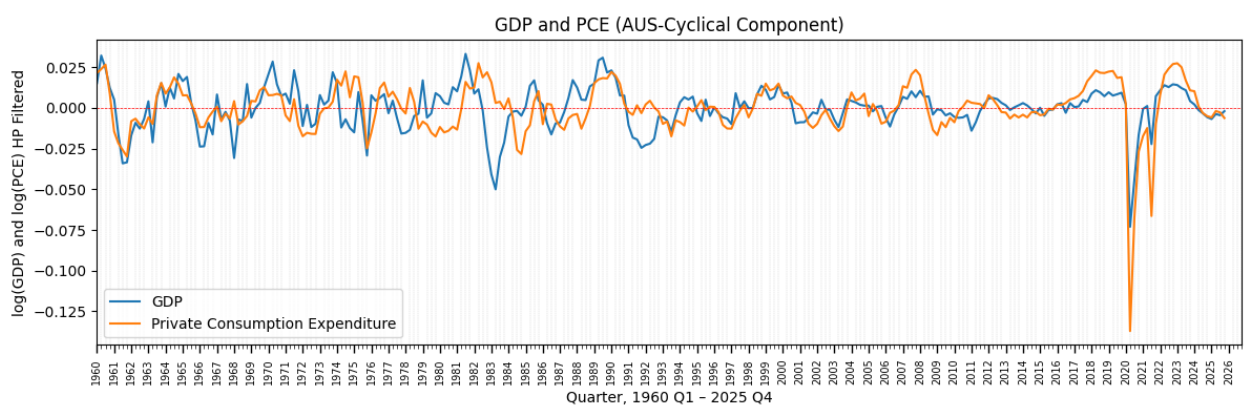

Figure 2. Cyclical Components of GDP and PCE – Australia

Notes: Both series are expressed in log terms and HP-filtered, $\lambda = 1600$.

1.2

	U.S.	Australia
$\sigma(GDP)$	1.5%	1.34%
$\sigma(PCE)$	1.356%	1.565%
Var (GDP)	0.000225	0.000179
Var (PCE)	0.000184	0.000245
Corr(C, GDP)	0.879	0.571
$Corr(GDP_{AUS},\ GDP_{US})$	—	0.382
$Corr(PCE_{AUS},\ GDP_{US})$	—	0.262
Period	1960:Q1 – 2025:Q4	1960:Q1 – 2025:Q4
Observations	264	264

Table 1. Business Cycle Moments: a comparison between the U.S. and Australia

Notes: Both series are expressed in log terms and HP-filtered, $\lambda = 1600$.

1.3 Discussion

Figure 1 plots the quarterly HP-filtered cycles of real GDP and real personal consumption expenditures (PCE) for the United States from 1960Q1 to 2025Q4. Figure 2 reports the analogous series for Australia. The data source for the original real GDP and real Personal/Private consumption expenditure series is FRED, Federal Reserve Bank of St. Louis.

A comparison between the two figures allows us to notice the following four facts. First, under HP filter trend decomposition, the U.S business cycle looks more persistent than the Australian one, as the oscillations around the trend (dashed red line) seem more frequent for the former than for the latter. Second, the dynamics of real personal consumption are more aligned with GDP in the U.S., while in the Australian case, one can easily notice moments of complete misalignment, such as those occurring in 1974-1975 and from 1979 to 1993. Third, both economies show a notably more contained cyclical deviation from the trend from the mid-1980s onwards, abstracting from the COVID-19 episode. This aligns with the Great Moderation, the well-documented period of declining macroeconomic volatility associated, among other factors, with the diffusion of inflation targeting regimes across advanced economies. Fourth, the two countries experienced markedly different dynamics during the two most recent major global downturns: the 2008 financial crisis and the COVID-19 recession, which began in early 2020. During the 2008–09 crisis, U.S. real GDP fell approximately 3% below its trend level, while Australia remained above trend throughout the episode, registering only a mild and short-lived cyclical contraction in the aftermath. The pattern broadly reversed during COVID-19: while both countries experienced a comparable initial decline in output relative to trend, Australian real consumption contracted far more sharply in the same period. Moreover, Australia recorded a further deterioration in both output and consumption during 2021, a secondary contraction with no counterpart in the

U.S. data. However, a closer look at both economies' business cycle moments offers additional insights. Table 1 reports the key moments of the four series shown in Figure 1 and Figure 2, and a few remarks are in order. First, the comparison of cyclical real GDP standard deviations suggests that, in a typical quarter, U.S. output deviates slightly more from its trend than its Australian counterpart. The opposite holds for real personal consumption expenditures, where Australia displays greater cyclical volatility than the United States. Second, the Australian consumption cyclical volatility is higher compared to its output volatility, in contrast to the well-established empirical regularity observed across most advanced economies, including the United States. Third, as noted earlier, the first correlation in Table 1 confirms that output and consumption dynamics in the U.S. are more aligned than in Australia. Lastly, both Australian consumption and output display very little co-movement with U.S. business cycle, suggesting that the south-eastern economy may depend on other global economic actor like China.

2 Local Projections and Macroeconomic Shocks

2.1 Replication

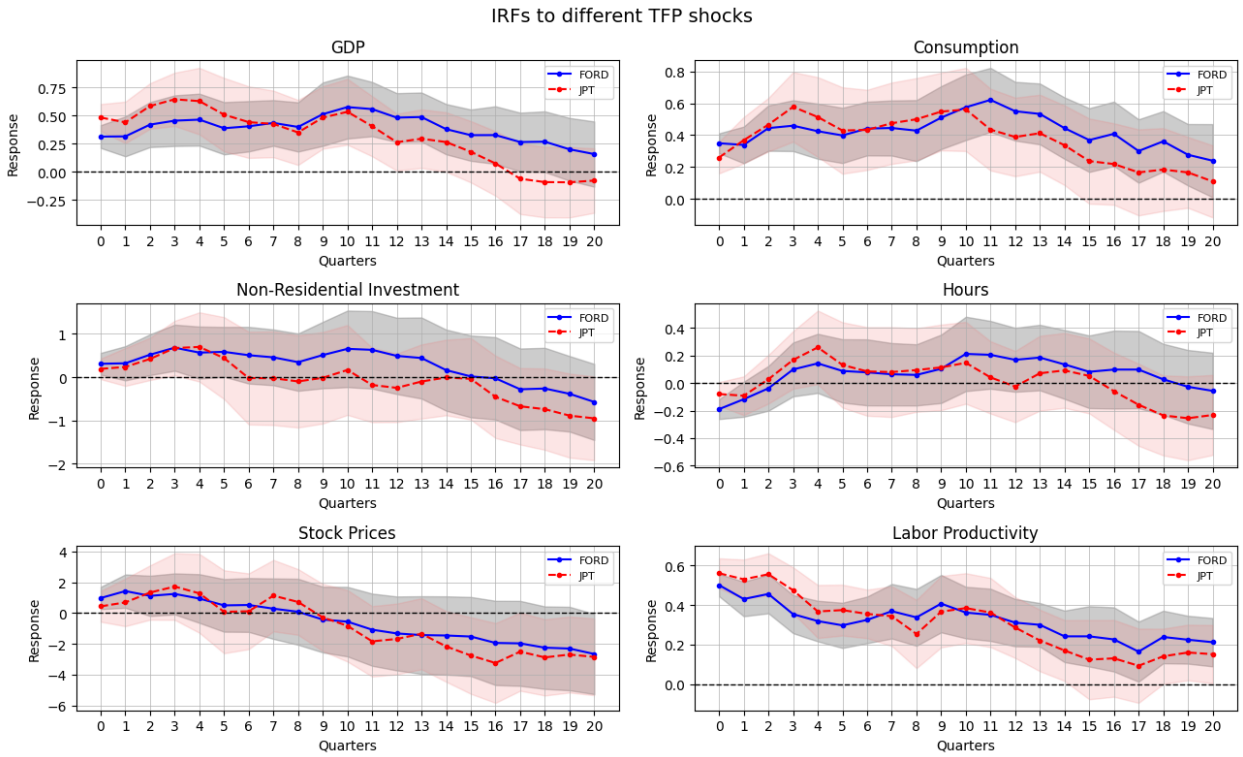

Figure 3. Impulse Response Functions to TFP shocks–United States

Notes: Blue lines with circles and grey bands: Francis et al. (2014) (FORD) TFP shock, *1949Q2-2009Q4*; Dashed red lines with circles and light-coral bands: Justiniano, Primiceri, and Tambalotti (2011) (JPT) TFP shock,*1954Q3-2009Q1*. Responses are estimated over a 20-quarter horizon using Jordà Local Projections; 90% Confidence Intervals and Newey-West standard errors.

Figure 3 replicates Fig.9 (ch.5.4) of Ramey (2016). It shows the estimated impulse response functions for two technology shocks on labor productivity, real per capita GDP, consumption, stock prices, non-residential investment, and per capita hours worked for 20 quarters ahead. The source of the original data is `Technology_data.xlsx`.
The replication follows the same Local Projection specification in Ramey (2016), which is equivalent to:[1]:

$$z_{t+h} = \alpha_h + \beta_h shock_t + \gamma(L) shock_{t-1} + \delta_h(L) z_{t-1} + \zeta_h(L) \boldsymbol{x_{t-1}} + \eta_h t + \theta_h t^2 + \varepsilon_{t+h}; \quad h = 0, 1, \ldots, 20. \tag{1}$$

where: z_{t+h} is the outcome variable in log terms, $shock_t$ is the standardized shock; $\boldsymbol{x_{t-1}}$ is a set of control variables in log-terms; t and t^2 represent a quadratic trend term; and ε_{t+h} is a serially correlated error term[2]. The number of lags for each lagged variable on the right-hand side is 2. The set of controls $\boldsymbol{x_{t-1}}$ includes, for each estimated I.R.F. log real GDP per capita, log real stock prices per capita, log labor productivity (equal to real GDP divided by total hours worked)[3]. The shocks considered are:

- The **FORD Total Factor Productivity** shock from Francis et al. (2014).
- The **JPT Total Factor Productivity** shock from Justiniano, Primiceri, and Tambalotti (2011).

The effects following a one standard deviation increase in the two technology shocks in Figure 3 suggest that:

- **Real per capita GDP** significantly increases on impact by 0.30% from the baseline for the FORD shock and 0.5% for the JPT one. The significance lasts for 3 years (JPT) and 4 years (FORD);
- **Real per capita Consumption** increases by 0.3% (FORD) and 0.3% (JPT) on impact, reaching a peak after 3 (JPT) and 11 quarters ahead;
- **Real per capita Non-Residential Investment** effects are significant only for the FORD shock after three years from impact;
- **Per Capita Hours Worked** responds significantly to the impact by a 0.2% drop for the FORD shock, while the effects are never significant for the JPT shock;
- **Real per capita Stock Prices** responds positively (1%) and significantly for two quarters after the impact of the FORD shock, and negatively after fifteen quarters after the impact of the JPT shock;

[1]The specification here mentioned is more specific in the notation for the lags of control variables and is not different from the one in Ramey (2016).

[2]For this reason, estimates' standard errors are computed by means of the Newey-West correction.

[3]These three variables are included altogether in the set of controls whenever none of these is the outcome variable to avoid collinearity

- **Labor Productivity** responds significantly with ≈0.5% increase for both shocks. The significant positive effects last 14 quarters for the JPT shock and for the whole considered forecast horizon for the FORD one.

2.2 Empirical Analysis

Research Question An important area of interest in macroeconomics has always regarded the effects of monetary policy on real economic activity, and in particular on industrial activity. How does monetary policy transmit to the latter? Are the different margins of adjustment of the manufacturing sector affected in the same way by monetary policy, which clearly represents a demand shock?

When an unexpected monetary policy tightening occurs, firms may face adjustment decisions along two distinct margins. Adjustments along the intensive margin amount to decide how much of their existing installed capacity (i.e. capital) to actually use. This margin is typically very flexible to regulate. In the case of monetary tightening, the firm would slow its plants without an irreversible commitment. A different story involves adjustments on the extensive margin, which is related to the decision on how many workers to employ. This channel is slower and more costly because hiring and firing entail search costs. Firms are therefore reluctant to cut employment until they are confident the shock is persistent.

This anecdotal prior would have a clear empirical prediction: Capacity utilization should respond earlier than manufacturing employment after a demand shock. However, nothing guarantees that the transmission of such a demand shock is symmetric across the direction of the shock.

The scope of this concise empirical analysis is to (i) document the differential adjustment timing along the two outlined margins after a monetary policy shock and (ii) assess empirically whether the effects of unexpected monetary tightening and easing are symmetric in magnitude and persistence.

Shock and Outcome Variables To this end, I estimate local projections à la Jordà (2005) using the narrative monetary policy shock of C. D. Romer and D. H. Romer (2004), extended to 2007 by Wieland and Yang (2020)[4], on monthly US manufacturing data over the period 1972m1–2007m12. Capacity utilization in manufacturing (`MCUMFN`) proxies the intensive margin: it measures the fraction of installed capital actually operated, which firms can adjust immediately and costlessly. Manufacturing employment (`MANEMP`) proxies the extensive margin: it captures the number of workers employed, an adjustment that is slower and more costly due to hiring and firing frictions. Both series are obtained from FRED, Federal Reserve Bank of St. Louis.

Data Transformations All outcome variables enter the local projections in log levels. Specifically, I construct $\log(\text{MCUMFN}_t)$ and $\log(\text{MANEMP}_t)$ by taking the natural logarithm of each original series. This transformation has two advantages. First, it ensures that the

[4]The extended shock series is obtained from Ramey (2016)'s dataset `Monetarydat.xlsx`, sheet `Monthlydat6907`.

impulse response coefficients $\hat{\beta}^h$ are interpretable as approximate percentage deviations of the outcome variable from its level in the counterfactual[5]. Second, the log transformation compresses the variance of both series and reduces the influence of outliers. The monetary policy shock ε_t^{RR} enters in levels -it is already expressed in percentage points of the unexpected change in the Federal Funds Rate and requires no further transformation. External control variables[6] include the monthly unemployment rate (UNEMP), the log-levels of the Consumer Price Index (LCPI), and the Federal Funds Rate (FFR) following Ramey (2016).

Specification For the average Impulse Response Functions, I estimate the following long-differences local projection for each outcome $y \in \{\log(\text{MCUMFN}), \log(\text{MANEMP})\}$ and each horizon $h = 0, 1, \ldots, H$: I estimate the following:

$$y_{t+h} - y_{t-1} = \alpha^h + \beta^h \varepsilon_t^{RR} + \sum_{k=1}^{p} \gamma_k^h y_{t-k} + \sum_{k=1}^{p} \boldsymbol{\delta}_k^h \mathbf{X}_{t-k} + u_{t+h} \tag{2}$$

where $\mathbf{X}_{t-k} = [\text{FFR}_{t-k}, \text{CPI}_{t-k}, \text{UNEMP}_{t-k}]'$ collects the external controls, $p = 2$ following Ramey (2016) in a similar exercise.

For the asymmetric effects, I estimate:

$$y_{t+h} - y_{t-1} = \alpha^h + \beta^{+,h} \varepsilon_t^{+} + \beta^{-,h} \varepsilon_t^{-} + \sum_{k=1}^{p} \gamma_k^h y_{t-k} + \sum_{k=1}^{p} \boldsymbol{\delta}_k^h \mathbf{X}_{t-k} + u_{t+h} \tag{3}$$

where $\varepsilon_t^{+} = \max\{\varepsilon_t^{RR}, 0\}$ and $\varepsilon_t^{-} = \max\{-\varepsilon_t^{RR}, 0\}$ decompose the shock into its tightening and easing components respectively, both expressed as positive quantities. The sequences $\{\hat{\beta}^{+,h}\}$ and $\{\hat{\beta}^{-,h}\}$ trace the cumulative Impulse Responses to a one percentage point unexpected tightening and easing, respectively. Asymmetry is present if $\hat{\beta}^{+,h} \neq \hat{\beta}^{-,h}$ for some h. Controls are the same for both specifications.

Results and Interpretation Figure 4 reports the cumulative impulse responses of manufacturing employment and capacity utilization to a one percentage point increase in the C. D. Romer and D. H. Romer (2004), estimated via equation (2). The shaded area represents 90% confidence bands based on HAC Newey-West standard errors.

The results are consistent with the anecdotal prior outlined above. A one percentage point unexpected monetary tightening generates a negative and persistent response in both variables, but with a very different timing across the two margins. In particular, Capacity utilization responds immediately: the IRF turns negative within the first 3-4 months, reaches a first trough of approximately -1.5% around m12, and a second one of approximately -2% around m24, before gradually recovering. Manufacturing employment, by contrast, remains positive

[5]The specification will show that in this case, I will focus on cumulative changes from the month preceding the shock, a specification usually referred to as "Long-Differences".

[6]The source of this data is Ramey (2016)'s dataset `Monetarydat.xlsx`, sheet `Monthlydat6907`.

for the first 6 months as firms do not immediately cut their workforce, and only begins declining around m8, reaching a trough of approximately -1.5% around m28-m30.

Figure 5 reports the impulse responses of manufacturing employment and capacity utilization to a one percentage point unexpected tightening ($\hat{\beta}^{+,h}$, blue) and easing ($\hat{\beta}^{-,h}$, red), estimated via equation (3).

The results reveal a clear asymmetry in the transmission of monetary policy across both margins. Looking first at manufacturing employment, an unexpected tightening generates a large and persistent contraction: the IRF reaches a trough of approximately -3.5% around m28-m30 and does not recover within the 36-month horizon. On the other hand, an unexpected easing generates a statistically significant negative response, which may be consistent with the firm interpreting this policy as a signal of deteriorating economic conditions. Nonetheless, employment does not exhibit any significant change after m6. These results suggest that the effects of monetary policy on the extensive margin are mainly driven by tightening episodes.

Focusing on capacity utilization, the asymmetry is pronounced as in the extensive counterpart discussed. A one percentage point tightening reduces capacity utilization by approximately -4% at the trough around m24-m26, with confidence bands that exclude zero in the medium term. The easing shock again generates a similar dynamic seen for employment, only more pronounced in the magnitude[7], with no significant persistent change over time.

Taken together, these results deliver the second main finding of this analysis: monetary policy operates asymmetrically along both margins of the manufacturing sector. Tightening generates large, significant, and persistent contractions in both capacity utilization and employment. Easing, by contrast, fails to produce a symmetric stimulus. These findings are consistent with the idea, recently documented by Jordà, Singh, and Taylor (2024) in a long-run oriented analysis, that central banks cannot manipulate the supply side of the economy by exploiting loose policy.

[7]This magnitude effect may again be consistent with the story of capacity utilization being more flexible than employment.

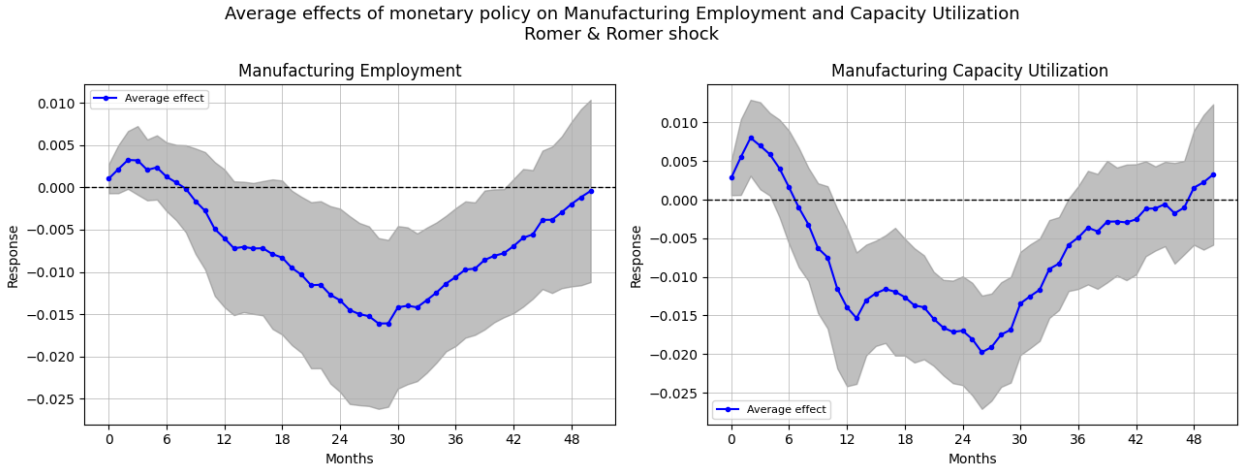

Figure 4. Impulse Response Functions to Monetary Policy shock–United States

Notes: Blue lines with circles and grey bands: C. D. Romer and D. H. Romer (2004) monetary policy shock, *1972m2-2007m12*; Cumulative Impulse Responses are estimated over a 36-month horizon using Jordà Local Projections in Long-Differences; 90% Confidence Intervals and Newey-West standard errors.

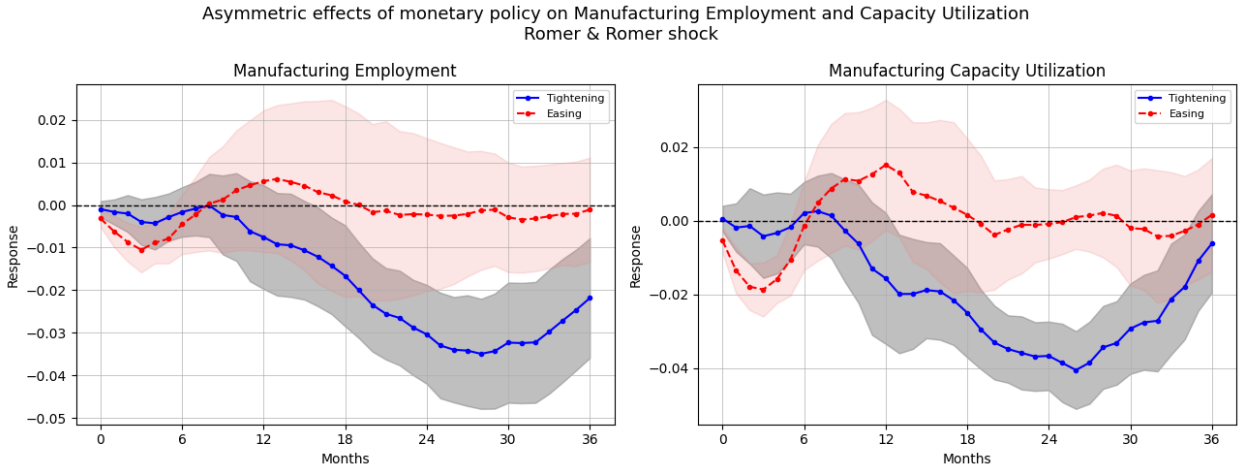

Figure 5. Impulse Response Functions to tightening and easing monetary policy shock–United States

Notes: Blue lines with circles and grey bands: C. D. Romer and D. H. Romer (2004) tightening monetary policy shock; Dashed red lines with circles and light-coral bands easing monetary policy shock. Sample:*1972m2-2007m12*. Cumulative Responses are estimated over a 36-month horizon using Jordà Local Projections in Long-Differences; 90% Confidence Intervals and Newey-West standard errors.

References

Francis, Neville et al. "A Flexible Finite-Horizon Alternative to Long-Run Restrictions with an Application to Technology Shocks". In: *The Review of Economics and Statistics* 96.4, pp. 638–647. ISSN: 00346535, 15309142. URL: http://www.jstor.org/stable/43554945 (visited on 03/31/2026).

Jordà, Òscar, Sanjay R. Singh, and Alan M. Taylor. "The Long-Run Effects of Monetary Policy". In: *The Review of Economics and Statistics*, pp. 1–49. ISSN: 0034-6535. DOI: 10.1162/rest_a_01527. eprint: https://direct.mit.edu/rest/article-pdf/doi/10.1162/rest_a_01527/2477344/rest_a_01527.pdf. URL: https://doi.org/10.1162/rest_a_01527.

Justiniano, Alejandro, Giorgio Primiceri, and Andrea Tambalotti. "Investment Shocks and the Relative Price of Investment". In: *Review of Economic Dynamics* 14.1, pp. 101–121. DOI: 10.1016/j.red.2010.08.004. URL: https://ideas.repec.org/a/red/issued/09-248.html.

Ramey, V.A. "Chapter 2 - Macroeconomic Shocks and Their Propagation". In: ed. by John B. Taylor and Harald Uhlig. Vol. 2. Handbook of Macroeconomics. Elsevier, pp. 71–162. DOI: https://doi.org/10.1016/bs.hesmac.2016.03.003. URL: https://www.sciencedirect.com/science/article/pii/S1574004816000045.

Romer, Christina D. and David H. Romer. "A New Measure of Monetary Shocks: Derivation and Implications". In: *American Economic Review* 94.4, pp. 1055–1084. DOI: 10.1257/0002828042002651. URL: https://www.aeaweb.org/articles?id=10.1257/0002828042002651.

Wieland, Johannes F and Mu-Jeung Yang. "Financial dampening". In: *Journal of Money, Credit and Banking* 52.1, pp. 79–113.